Supplementary Material

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APPENDIX A
PROOF OF ACHIEVABLE RATE

Considering $\mathbf{R}_{\mathbf{r_{u_m}}} = \mathbb{E}[\mathbf{r}_{u_m}\mathbf{r}_{u_m}^H]$ and $\mathbf{R}_m(\mathbf{Q}_{-j_m}, w_m)$ as the covariance matrix of the received signal at m^{th} UAV and the normalized covariance matrix of the noise plus interference at UAV u_m, respectively, we can obtain the achievable rate related to the link between IMD j_m and the UAV u_m, as follows:

$$\begin{aligned} & r_{j_m}(\mathbf{Q}, w_m) \\ = \;& w_m \log_2\left(\det\left(\mathbf{R}_{\mathbf{r_{u_m}}}\right)\right) - w_m \log_2\left(\det\left(\frac{\mathbf{R}_m}{\lambda}\right)\right) \\ \overset{(a)}{=} \;& w_m \log_2\left(\det\left(\bar{\mathbf{H}}_{j_m,u_m}\mathbf{Q}_{j_m}\bar{\mathbf{H}}_{j_m,u_m}^H + \frac{\mathbf{R}_m}{\lambda}\right)\right) \\ & - w_m \log_2\left(\det\left(\frac{\mathbf{R}_m}{\lambda}\right)\right), \end{aligned}$$

where we drop $(\mathbf{Q}_{-j_m}, w_m)$ from $\mathbf{R}_m(\mathbf{Q}_{-j_m}, w_m)$ for simplicity in notations. Moreover, (a) comes from the following formula:

$$\begin{aligned} \mathbf{R}_{\mathbf{r_{u_m}}} &= \mathbb{E}\left[\mathbf{r}_{u_m}\mathbf{r}_{u_m}^H\right] = \mathbb{E}\left[\left(\bar{\mathbf{H}}_{j_m,u_m}\mathbf{x}_{j_m} + \mathbf{n}_m\right)\right. \\ &\left.\left(\bar{\mathbf{H}}_{j_m,u_m}\mathbf{x}_{j_m} + \mathbf{n}_m\right)^H\right] = \bar{\mathbf{H}}_{j_m,u_m}\mathbb{E}\left[\mathbf{x}_{j_m}\mathbf{x}_{j_m}^H\right]\bar{\mathbf{H}}_{j_m,u_m}^H \\ &+ \bar{\mathbf{H}}_{j_m,u_m}\mathbb{E}\left[\mathbf{x}_{j_m}\mathbf{n}_m^H\right] + \bar{\mathbf{H}}_{j_m,u_m}^H\mathbb{E}\left[\mathbf{n}_m\mathbf{x}_{j_m}^H\right] \\ &+ \mathbb{E}\left[\mathbf{n}_m\mathbf{n}_m^H\right] = \bar{\mathbf{H}}_{j_m,u_m}\mathbf{Q}_{j_m}\bar{\mathbf{H}}_{j_m,u_m}^H + \frac{\mathbf{R}_m}{\lambda}, \end{aligned}$$

where $\lambda \triangleq \frac{1}{N_0 w_m}$ and $\mathbf{n}_m$ denotes the noise plus interference vector received at UAV u_m. Then, the achievable rate $r_{j_m}(\mathbf{Q}, w_m)$ is rewritten as

$$\begin{aligned} & r_{j_m}(\mathbf{Q}, w_m) \\ = \;& w_m \log_2\left(\det\left[\left(\bar{\mathbf{H}}_{j_m,u_m}\mathbf{Q}_{j_m}\bar{\mathbf{H}}_{j_m,u_m}^H + \frac{\mathbf{R}_m}{\lambda}\right)\lambda\mathbf{R}_m^{-1}\right]\right) \\ = \;& w_m \log_2\left(\det\left[\lambda\bar{\mathbf{H}}_{j_m,u_m}\mathbf{Q}_{j_m}\bar{\mathbf{H}}_{j_m,u_m}^H\mathbf{R}_m^{-1} + \mathbf{I}_{A_m}\right]\right) \\ \overset{(b)}{=} \;& w_m \log_2\left(\det\left[\mathbf{I}_{A_{j_m}} + \lambda\bar{\mathbf{H}}_{j_m,u_m}^H\mathbf{R}_m^{-1}\bar{\mathbf{H}}_{j_m,u_m}\mathbf{Q}_{j_m}\right]\right), \end{aligned}$$

where (b) comes from the formula $\det(\mathbf{AB} + \mathbf{I}_{A_m}) = \det(\mathbf{BA} + \mathbf{I}_{A_{j_m}})$, in which $\mathbf{A} = \bar{\mathbf{H}}\mathbf{Q}_{j_m} \in \mathbb{C}^{A_m \times A_{j_m}}$ and $\mathbf{B} = \bar{\mathbf{H}}^H\mathbf{R}_m^{-1} \in \mathbb{C}^{A_{j_m} \times A_m}$.

APPENDIX B
PROOF OF THE OPTIMALITY OF THE SMOOTHED ℓ_p-norm FUNCTION

At first, we should prove that $\limsup_{\eta \searrow 0} \mathcal{P}(\eta) \subset \mathcal{P}(0)$. Based on the results in [1, Page 152], $\limsup_{\eta \searrow 0} \mathcal{P}(\eta)$ can be defined as follows:

$$\begin{aligned} \limsup_{\eta \searrow 0} \mathcal{P}(\eta) &:= \bigcup_{\eta(\upsilon)\searrow 0} \limsup_{\eta \to \infty} \mathcal{P}(\eta(\upsilon)) \\ &= \left\{\overline{\boldsymbol{\nu}} | \exists\, \eta(\upsilon) \searrow 0, \exists\, \boldsymbol{\nu}(\upsilon) \to \overline{\boldsymbol{\nu}}\right\}, \end{aligned} \tag{1}$$

where $\boldsymbol{\nu}(\upsilon) \in \mathcal{P}(\eta(\upsilon))$. Hence, for any $\overline{\boldsymbol{\nu}} \in \limsup_{\eta \searrow 0} \mathcal{P}(\eta)$, $\boldsymbol{\nu}(\upsilon) \in \mathcal{P}(\eta(\upsilon))$ exists so that $\boldsymbol{\nu}(\upsilon) \to \overline{\boldsymbol{\nu}}$ and $\eta(\upsilon) \searrow 0$. In order to derive (39), we should prove that $\overline{\boldsymbol{\nu}} \in \mathcal{P}(0)$. More precisely, $\boldsymbol{\nu}(\upsilon) \in \mathcal{P}(\eta(\upsilon))$ denotes that

$$0 \in \nabla_{\boldsymbol{\nu}} f(\boldsymbol{\nu}(\upsilon); \eta(\upsilon)) + \mathcal{L}_{\boldsymbol{\mathcal{Z}}}(\boldsymbol{\nu}(\upsilon)). \tag{2}$$

Since $f(\boldsymbol{\nu}; \eta)$ is continuously differentiable relative to $\boldsymbol{\nu}$ and η, we have

$$\begin{aligned} \lim_{\upsilon \to \infty} \nabla_{\boldsymbol{\nu}} f(\boldsymbol{\nu}(\upsilon); \eta(\upsilon)) &= \lim_{\boldsymbol{\nu}(\upsilon) \to \overline{\boldsymbol{\nu}}} \lim_{\eta(\upsilon)\searrow 0} \nabla_{\boldsymbol{\nu}} f(\boldsymbol{\nu}(\upsilon); \eta(\upsilon)) \\ &= \nabla_{\boldsymbol{\nu}} f(\overline{\boldsymbol{\nu}}; 0). \end{aligned} \tag{3}$$

Moreover, according to the results in [1, Proposition 6.6], we have

$$\limsup_{\boldsymbol{\nu}(\upsilon) \to \overline{\boldsymbol{\nu}}} \mathcal{L}_{\boldsymbol{\mathcal{Z}}}(\boldsymbol{\nu}(\upsilon)) = \mathcal{L}_{\boldsymbol{\mathcal{Z}}}(\overline{\boldsymbol{\nu}}). \tag{4}$$

Specifically, if $\boldsymbol{\nu}(\upsilon) \to \overline{\boldsymbol{\nu}}, x(\upsilon) \in \mathcal{L}_{\boldsymbol{\mathcal{Z}}}(\boldsymbol{\nu}(\upsilon))$ and $x(\upsilon) \to x$, then $x \in \mathcal{L}_{\boldsymbol{\mathcal{Z}}}(\overline{\boldsymbol{\nu}})$ meaning that the set $\{(\boldsymbol{\nu}, x) | x \in \mathcal{L}_{\boldsymbol{\mathcal{Z}}}(\boldsymbol{\nu})\}$ relative to $\boldsymbol{\mathcal{Z}} \times \mathbb{R}$ is closed. According to (3) and (4) and considering $\upsilon \to \infty$ in (2), we thus prove $\limsup_{\eta \searrow 0} \mathcal{P}(\eta) \subset \mathcal{P}(0)$. Based on [2, Theorem 4], proof of (39) is completed.

APPENDIX C
OBTAINING A CONVEX UPPER BOUND FOR SMOOTHED ℓ_p-NORM OBJECTIVE FUNTION

It is clear that the adjustment regularization parameter η prevents the infinity of the objective function when some

values of $\boldsymbol{\nu}(\upsilon)$ are equal to zero in the iterations. Then, the approximation error is defined as:

$$\begin{aligned} & f\left(\boldsymbol{\nu}^t, \boldsymbol{\nu}^e; \eta\right) - \mathcal{D}\left(\boldsymbol{\nu}; \xi\left(\upsilon\right)\right) = \\ & \sum_{j_m \in \boldsymbol{\chi}} \left(\left(\nu_{j_m}^t\right)^2 + \eta^2\right)^{p/2} + \sum_{j_m \in \boldsymbol{\chi}} \left(\left(\nu_{j_m}^e\right)^2 + \eta^2\right)^{p/2} \\ & - \sum_{j_m \in \boldsymbol{\chi}} \xi_{j_m}^t\left(\upsilon\right)\left(\nu_{j_m}^t\right)^2 - \sum_{j_m \in \boldsymbol{\chi}} \xi_{j_m}^e\left(\upsilon\right)\left(\nu_{j_m}^e\right)^2 \\ & = \left(\sum_{j_m \in \boldsymbol{\chi}} \left(\left(\nu_{j_m}^t\right)^2 + \eta^2\right)^{p/2} - \sum_{j_m \in \boldsymbol{\chi}} \xi_{j_m}^t\left(\upsilon\right)\left(\nu_{j_m}^t\right)^2\right) \\ & + \left(\sum_{j_m \in \boldsymbol{\chi}} \left(\left(\nu_{j_m}^e\right)^2 + \eta^2\right)^{p/2} - \sum_{j_m \in \boldsymbol{\chi}} \xi_{j_m}^e\left(\upsilon\right)\left(\nu_{j_m}^e\right)^2\right). \end{aligned} \tag{5}$$

For ease of mathematical expressions, we define $\mathcal{G}(\tau) \triangleq \left(\tau + \eta^2\right)^{p/2}$ with $\tau \geqslant 0$, which is strictly concave. In this regard, (5) can be rewritten as follows:

$$\begin{aligned} & f\left(\boldsymbol{\nu}^t, \boldsymbol{\nu}^e; \eta\right) - \mathcal{D}\left(\boldsymbol{\nu}; \xi\left(\upsilon\right)\right) = \\ & \left(\sum_{j_m \in \boldsymbol{\chi}} \mathcal{G}\left(\left(\nu_{j_m}^t\right)^2\right) - \mathcal{G}'\left(\left(\nu_{j_m}^t\left(\upsilon\right)\right)^2\right)\left(\nu_{j_m}^t\right)^2\right) \\ & + \left(\sum_{j_m \in \boldsymbol{\chi}} \mathcal{G}\left(\left(\nu_{j_m}^e\right)^2\right) - \mathcal{G}'\left(\left(\nu_{j_m}^e\left(\upsilon\right)\right)^2\right)\left(\nu_{j_m}^e\right)^2\right), \end{aligned} \tag{6}$$

where its maximum value is achieved at $\boldsymbol{\nu} = \boldsymbol{\nu}(\upsilon)$. More precisely, error function (6) is the sum of functions in the form of $\mathcal{B}(\tau)$, defined as $\mathcal{B}(\tau) = \mathcal{G}(\tau) - \mathcal{G}'(\tau)\tau$, where $\tau \geqslant 0$. In this regard, we should prove that the function $\mathcal{B}(\tau)$ with $\tau \geqslant 0$ achieves its maximum value at $\nu = \boldsymbol{\nu}(\upsilon)$, which is straightforward by taking a derivative of $\mathcal{B}(\tau)$; thus, the expression $\mathcal{B}'(\tau^{\upsilon}) = 0$ is achieved as the result of differentiation, and the proof is completed.

APPENDIX D
PROOF OF DUALITY OF EQUATIONS 35 AND 40

Let denote $\boldsymbol{\nu}(\upsilon+1)$ as the best solution obtained in υ^{th} iteration for the upper bound function $\mathcal{D}\left(\boldsymbol{\nu}; \xi(\upsilon)\right)$, i.e.,

$$\boldsymbol{\nu}\left(\upsilon+1\right) := \arg\min_{\boldsymbol{\nu}} \mathcal{D}\left(\boldsymbol{\nu}; \xi\left(\upsilon\right)\right). \tag{7}$$

Then, we have

$$\begin{aligned} f\left(\boldsymbol{\nu}\left(\upsilon+1\right); \eta\right) &= f\left(\boldsymbol{\nu}\left(\upsilon+1\right); \eta\right) - \mathcal{D}\left(\boldsymbol{\nu}\left(\upsilon+1\right); \xi\left(\upsilon\right)\right) \\ &\quad + \mathcal{D}\left(\boldsymbol{\nu}\left(\upsilon+1\right); \xi\left(\upsilon\right)\right) \\ &\leqslant f\left(\boldsymbol{\nu}\left(\upsilon\right); \eta\right) - \mathcal{D}\left(\boldsymbol{\nu}\left(\upsilon\right); \xi\left(\upsilon\right)\right) \\ &\quad + \mathcal{D}\left(\boldsymbol{\nu}\left(\upsilon+1\right); \xi\left(\upsilon\right)\right) \\ &\leqslant f\left(\boldsymbol{\nu}\left(\upsilon\right); \eta\right) - \mathcal{D}\left(\boldsymbol{\nu}\left(\upsilon\right); \xi\left(\upsilon\right)\right) \\ &\quad + \mathcal{D}\left(\boldsymbol{\nu}\left(\upsilon\right); \xi\left(\upsilon\right)\right) \\ &\leqslant f\left(\boldsymbol{\nu}\left(\upsilon\right); \eta\right), \end{aligned} \tag{8}$$

where the first and the second inequalities are derived based on Proposition 1 and (7), respectively. Based on (8), it is concluded that the point minimizing the upper bound (40) can also minimize the (35), and the proof is completed.

APPENDIX E
PROOF OF FIRST-ORDER TAYLOR SERIES OF MATRIX FUNCTIONS

As previously mentioned, $r_{j_m}^{+}(\mathbf{Q}_{-j_m}, w_m)$-$r_{j_m}^{-}(\mathbf{Q}_{-j_m}, w_m)$ is non-convex. To find a convex upper bound for the $r_{j_m}^{-}(\mathbf{Q}_{-j_m}, w_m)$ in Eq. (49), we utilize the first-order Taylor series of real function $r_{j_m}^{-}(\mathbf{Q}_{-j_m}, w_m)$ with respect to real vectors, expressed as

$$\begin{aligned} & r_{j_m}^{-}(\mathbf{Q}_{-j_m}, w_m) = r_{j_m}^{-}(\mathbf{Q}_{j_m}(\upsilon), w_m(\upsilon)) \\ & + \left(\mathrm{vec}\left(\nabla_{\mathbf{Q}^* j_m} r_{j_m}^{-}(\mathbf{Q}_{-j_m}(\upsilon), w_m)\right)\right)\mathrm{vec}\left(\mathbf{Q}_{j_m} - \mathbf{Q}_{j_m}(\upsilon)\right) \\ & + \frac{\partial r_{j_m}^{-}(\mathbf{Q}_{-j_m}, w_m(\upsilon))}{\partial w_m^*}\left(w_m - w_m(\upsilon)\right), \end{aligned} \tag{9}$$

where $\mathrm{vec}(\mathbf{A})$ denotes the vectorization of matrix $\mathbf{A}$, in which the column entries of the matrix $\mathbf{A}$ is put on top of one another to make $\mathrm{vec}(\mathbf{A})$. To calculate $\mathrm{vec}\left(\nabla_{\mathbf{Q}-j_m} r_{j_m}^{-}(\mathbf{Q}_{-j_m}, w_m)\right)$, we should compute $\frac{\partial r_{j_m}^{-}(\mathbf{Q}_{-j_m}, w_m)}{\partial(\mathrm{vec}^T(\mathbf{Q}_{j_m}))}$ as follows:

$$\begin{aligned} & \partial r_{j_m}^{-}(\mathbf{Q}_{-j_m}, w_m) \\ & \overset{(c)}{=} 2\sum_{j'=1}^{j-1} w_m \mathrm{Re}\left(\mathrm{tr}\left[\mathbf{R}_m^{-1}(\mathbf{Q}_{-j'_m}, w_m)\partial\mathbf{R}_m(\mathbf{Q}_{-j'_m}, w_m)\right]\right) \\ & = 2\sum_{j'=1}^{j-1} w_m \mathrm{Re}\left(\mathrm{tr}\left[\mathbf{R}_m^{-1}(\mathbf{Q}_{-j'_m}, w_m)\mathbf{H}_{j'_m,u_m}\partial\mathbf{Q}_{j'_m}\mathbf{H}_{j'_m,u_m}^H\right]\right) \\ & = 2\sum_{j'=1}^{j-1} w_m \mathrm{Re}\left(\mathrm{tr}\left[\mathbf{H}_{j'_m,u_m}^H\mathbf{R}_m^{-1}(\mathbf{Q}_{-j'_m}, w_m)\mathbf{H}_{j'_m,u_m}\partial\mathbf{Q}_{j'_m}\right]\right) \\ & = 2\sum_{j'=1}^{j-1} w_m \mathrm{Re}\left(\mathrm{tr}\left[\mathrm{vec}^T\left(\mathbf{H}_{j'_m,u_m}^H\mathbf{R}_m^{-1}(\mathbf{Q}_{-j'_m}, w_m)\mathbf{H}_{j'_m,u_m}\right)^T\right.\right. \\ & \left.\left.\mathrm{vec}\left(\partial\mathbf{Q}_{j'_m}\right)\right]\right) = 2\sum_{j'=1}^{j-1} w_m \mathrm{Re}\left(\mathrm{vec}^T\left(\mathbf{H}_{j'_m,u_m}^H\mathbf{R}_m^{-1}(\mathbf{Q}_{-j'_m}, w_m)\right.\right. \\ & \left.\left.\mathbf{H}_{j'_m,u_m}\right)^T\right)\partial\left(\mathrm{vec}\left(\mathbf{Q}_{j'_m}\right)\right), \end{aligned}$$

where (c) comes from the formula $d\ \ln|(det(\mathbf{X}))| = \mathrm{tr}(\mathbf{X}^{-1}d\mathbf{X} + \mathbf{X}^{*-1}d\mathbf{X}^*) = 2\mathrm{Re}\left(\mathrm{tr}\left(\mathbf{X}^{-1}d\mathbf{X}\right)\right)$ by considering the fact that $\ln|(det(\mathbf{X}^*))| = \ln|(det(\mathbf{X}))|$. Accordingly, we obtain $\left(\frac{\partial r_{j_m}^{-}(\mathbf{Q}_{-j_m}, w_m)}{\partial \mathrm{vec}^T(\mathbf{Q}_{j_m})}\right)$ as follows:

$$\begin{aligned} & \frac{\partial r_{j_m}^{-}(\mathbf{Q}_{-j_m}, w_m)}{\partial \mathrm{vec}^T(\mathbf{Q}_{j_m})} \\ & = 2\sum_{j'=1}^{j-1} w_m \mathrm{Re}\left(\mathrm{vec}^T\left(\mathbf{H}_{j'_m,u_m}^H\mathbf{R}_m^{-1}(\mathbf{Q}_{-j'_m}, w_m)\mathbf{H}_{j'_m,u_m}\right)\right). \end{aligned} \tag{10}$$

Then, $\left(\mathrm{vec}\left(\nabla_{\mathbf{Q}j_m{}^*} r_{j_m}^{-}(\mathbf{Q}_{-j_m}(\upsilon), w_m)\right)\right)\mathrm{vec}\left(\mathbf{Q}_{j_m} - \mathbf{Q}_{j_m}(\upsilon)\right)$

is calculated as follows:

$$\begin{aligned}
&\left(\text{vec}\left(\nabla_{\mathbf{Q}^*-j_m} r^-_{j_m}(\mathbf{Q}_{-j_m}(\upsilon), w_m)\right)\right)\text{vec}\left(\mathbf{Q}_{j_m}-\mathbf{Q}_{j_m}(\upsilon)\right)\\
&=2w_m\sum_{j'=1}^{j-1}\text{Re}\left(\text{vec}^T\left(\mathbf{H}^H_{j'_m,u_m}\mathbf{R}^{-1}_m(\mathbf{Q}_{-j'_m},w_m)\mathbf{H}_{j'_m,u_m}\right)^*\right)\\
&\times\text{vec}(\mathbf{Q}_{j_m}-\mathbf{Q}_{j_m}(\upsilon))\\
&=2\sum_{j'=1}^{j-1}w_m\text{Re}\left(\left(\mathbf{H}^H_{j'_m,u_m}\mathbf{R}^{-1}_m(\mathbf{Q}_{-j'_m},w_m)\mathbf{H}_{j'_m,u_m}\right)^H\times\right.\\
&\left.(\mathbf{Q}_{j_m}-\mathbf{Q}_{j_m}(\upsilon))\right)\\
&=\sum_{j'=1}^{j-1}2w_m\left\langle\left(\mathbf{H}^H_{j'_m,u_m}\mathbf{R}^{-1}_m\mathbf{H}_{j'_m,u_m}\right),\mathbf{Q}_{j_m}-\mathbf{Q}_{j_m}(\upsilon)\right\rangle.
\end{aligned}\tag{11}$$

With a similar argument as in (10), we can compute $\frac{\partial r^-_{j_m}(\mathbf{Q}_{-j_m},w_m)}{\partial w_m}$ as follows:

$$\begin{aligned}
&\partial r^-_{j_m}(\mathbf{Q}_{-j_m},w_m)\\
&=2\sum_{j'=1}^{j-1}w_m\text{Re}\left(\text{tr}\left[\mathbf{R}^{-1}_m(\mathbf{Q}_{-j'_m},w_m)\partial\mathbf{R}_m(\mathbf{Q}_{-j'_m},w_m)\right]\right)\\
&+\partial w_m\frac{r^-_{j_m}(\mathbf{Q}_{-j_m},w_m(\upsilon))}{w_m(\upsilon)}\\
&=2\sum_{j'=1}^{j-1}w_m\text{Re}\left(\text{tr}\left[\mathbf{R}^{-1}_m(\mathbf{Q}_{-j'_m},w_m)\partial\lambda\mathbf{H}_{j'_m,u_m}\mathbf{Q}_{j'_m}\mathbf{H}^H_{j'_m,u_m}\right]\right)\\
&+\partial w_m\frac{r^-_{j_m}(\mathbf{Q}_{-j_m},w_m(\upsilon))}{w_m(\upsilon)}\\
&=\left(-2\lambda\sum_{j'=1}^{j-1}\text{Re}\left(\text{tr}\left[\mathbf{H}^H_{j'_m,u_m}\mathbf{R}^{-1}_m(\mathbf{Q}_{-j'_m},w_m)\mathbf{H}_{j'_m,u_m}\mathbf{Q}_{j'_m}\right]\right)\right.\\
&\left.+\frac{r^-_{j_m}(\mathbf{Q}_{-j_m},w_m(\upsilon))}{w_m(\upsilon)}\right)\partial w_m.
\end{aligned}$$

Accordingly, $\frac{\partial r^-_{j_m}(\mathbf{Q}_{-j_m},w_m)}{\partial w^*_m}(w_m-w_m(\upsilon))$ is obtained as

$$\begin{aligned}
&\frac{\partial r^-_{j_m}(\mathbf{Q}_{-j_m},w_m)}{\partial w^*_m}(w_m-w_m(\upsilon))\\
&=\left(-2\lambda\sum_{j'=1}^{j-1}\text{Re}\left(\text{tr}\left[\mathbf{H}^H_{j'_m,u_m}\mathbf{R}^{-1}_m(\mathbf{Q}_{-j'_m},w_m)\mathbf{H}_{j'_m,u_m}\mathbf{Q}_{j'_m}\right]\right)\right.\\
&\left.+\frac{r^-_{j_m}(\mathbf{Q}_{-j_m},w_m(\upsilon))}{w_m(\upsilon)}\right)(w_m-w_m(\upsilon)).
\end{aligned}\tag{12}$$

Substituting (10) and (12) into (9), $r^-_{j_m}(\mathbf{Q}_{-j_m},w_m)$ is computed and the proof is completed.

APPENDIX F
PROOF OF COMPUTATIONAL COMPLEXITY OF ALGORITHM 1

Regarding pseudo code in **Algorithm** 1, there is only one *"while"* loop (i.e., steps $1-20$) containing an inner *"for"* loop in steps $3-11$. Thus, the computational complexity of **Algorithm** 1 mainly depends on this loop. More precisely, the computational complexity of Algorithm 1 mainly relates to lines $4-6$ belonging to the *"for"* loop, and its value for line 4, 5, and 6 are obtained as $\mathcal{O}(t\Omega)$, $\mathcal{O}(t\Omega\hat{M})$, and $\mathcal{O}(t\Omega\hat{M})$, respectively. However, the complexity order of other lines can be neglected compared to the complexity order of the lines $4-6$. Accordingly, the total computational complexity of **Algorithm** 1 is calculated by adding the aforementioned complexity terms, i.e., $\mathcal{O}(t\Omega\hat{M}+t\Omega)$, or equivalently, $\mathcal{O}\left(t\Omega\hat{M}\right)$.

APPENDIX G
PROOF OF COMPUTATIONAL COMPLEXITY OF ALGORITHM 1

There is only a bottleneck line in complexity analysis of **Algorithm** 2 related to solving the problem $\mathscr{P}\mathbf{4}$ at line 1, where we employ the SCA algorithm. To obtain its computational complexity, we need to calculate the number of multiplications performed in each iteration of the Algorithm stemming from the three constraints $\mathbf{C1}'\backsim\mathbf{C3}'$ in subproblem $\mathscr{P}\mathbf{4}$, while the computational complexity of other constraints is low and can be neglected. As far as (**??**) exists in all constraints $\mathbf{C1}'\backsim\mathbf{C3}'$, we first focus on obtaining its computational complexity, which mainly depends on the complexity of the matrix determinant. Accordingly, its complexity is obtained of order $\mathcal{O}(\sum_{j_m\in\boldsymbol{\chi}_\ell}A^3_{j_m})$ that is the dominant term for computing the complexity. Furthermore, the complexity of (**??**) equals to $\mathcal{O}(|\boldsymbol{\chi}_\ell|)$. Based on the above argument, the computational complexity for constraints $\mathbf{C1}'\backsim\mathbf{C3}'$ are of order $\mathcal{O}\left(|\boldsymbol{\chi}_\ell|+\sum_{j_m\in\boldsymbol{\chi}_\ell}A^3_{j_m}\right)$, $\mathcal{O}\left(\sum_{j_m\in\boldsymbol{\chi}_\ell}A^3_{j_m}\right)$, and $\mathcal{O}\left(\sum_{j_m\in\boldsymbol{\chi}_\ell}A^3_{j_m}\right)$, respectively. Therefore, the total complexity of this step can be calculated by adding the aforementioned complexity terms, i.e., $\mathcal{O}\left(|\boldsymbol{\chi}_\ell|+7\sum_{j_m\in\boldsymbol{\chi}}A^3_{j_m}\right)$. On the other hand, $\mathscr{P}\mathbf{4}$ is solved using IPM in each implementation, which has a complexity of order $\mathcal{O}(\Theta^2_\ell\Xi_\ell)$ [**?**, Ch. 1], where Θ_ℓ denotes the number of the optimization variables, calculated as $\Theta_\ell=\sum_{j_m\in\boldsymbol{\chi}_\ell}A^2_{j_m}+2|\boldsymbol{\chi}_\ell|+\hat{M}$. In addition, Ξ_ℓ represents the number of constraints, calculated as $\Xi_\ell=6|\boldsymbol{\chi}_\ell|+\hat{M}+1$. Employing this method, we can claim that each case of problem $\mathscr{P}\mathbf{4}$ can be solved with the complexity of order $\mathcal{O}\left(\Theta^2_\ell\Xi_\ell\left(|\boldsymbol{\chi}_\ell|+7\sum_{j_m\in\boldsymbol{\chi}_\ell}A^3_{j_m}\right)\right)$. The computational complexity of other lines of Algorithm 2 is low and can be neglected. Therefore, the total computational complexity of $\mathscr{P}\mathbf{4}$ can be calculated by adding the aforementioned complexity terms, i.e.,

$$\mathcal{O}\left(\max\left\{\sum_{\ell=1}^{j}\left(\mathcal{F}_\ell\Theta^2_\ell\Xi_\ell\left(|\boldsymbol{\chi}_\ell|+7\sum_{j_m\in\boldsymbol{\chi}_\ell}A^3_{j_m}\right)\right)\right\}\right).$$

Assuming that all devices are equipped with the same number of antennas, i.e., $A_{j_m}=A$, $\forall j_m\in\boldsymbol{\chi}_{\ell'}$, we can simplify

the computational complexity of the overall algorithm as

$$\mathcal{O}\left(\max\left\{\sum_{\ell=1}^{L}\left(\mathcal{F}_\ell\Theta_\ell^2\Xi_\ell\left(|\boldsymbol{\chi}_\ell|+7\,|\boldsymbol{\chi}_\ell|\,A^3\right)\right)\right\}\right),$$

$$\text{or equivalently, } \mathcal{O}\left(A^7\sum_{l=1}^{L}\mathcal{F}_\ell|\boldsymbol{\chi}_\ell|^4\right).$$

APPENDIX H
CONVERGENCE ANALYSIS

In this subsection, we present the convergence analysis of Algorithms 1 related to subproblem $\mathscr{P}\mathbf{1}$. In this regard, we first define the first-order Markov chain process in Definition 1. Then, we prove that the 3D placement of $\hat{M}$ UAVs related to the population with Ω members in sequential iterations is a Markov chain process. Accordingly, the state, the state space, and the transition probability between two states/iterations are computed. Next, the martingale theory is introduced in Definition 2 to simplify the convergence of Algorithm 1. According to the Definition 2 and Lemma 2, we prove that Algorithm 1 converges to an optimal global value. Finally, we prove the convergence of Algorithm 2 related to subproblem $\mathscr{P}\mathbf{4}$ through Theorem 2.

Definition 1: Let denote $\{\sigma_i^t, t\geq 0\}$ as a discrete stochastic process related to i^{th} member of population $\{1,...,\Omega\}$ in t^{th} iteration. In this case, σ_i^t is the first-order Markov chain process if the following conditions are hold:

$$\begin{aligned}1)\ &\mathbb{E}\left[\sigma_i^{t+1}=s_{t+1}|\sigma_i^t=s_t,\sigma_i^{t-1}=s_{t-1},...,\sigma_i^0=s_0\right] \\ &=\mathbb{E}\left[\sigma_i^{t+1}=s_{t+1}|\sigma_i^t=s_t\right],\end{aligned}\tag{13}$$

$$2)\ \Pr\{\sigma_i^t=s_t,\sigma_i^{t-1}=s_{t-1},...,\sigma_i^0=s_0\}>0,\tag{14}$$

where $\{s_\ell|0\leq\ell\leq t+1\}$ refers to the set of values related to discrete states in the Markov chain process. In the proposed algorithm, each state of the process only depends on the previous state. More precisely, the parameter $\mathbf{g}_i^t$ is updated whenever the fitness function's value in the current state is lower than its value during the previous state. Accordingly, the 3D deployment of UAVs is considered the first-order Markov chain process. Let define the state, state-space, and other relevant concepts used to prove the global convergence of Algorithm 1.

The 3D positions of $\hat{M}$ UAVs related to i^{th} member of the population $\{1,...,\Omega\}$, denoted by $\mathbf{g}_i^t=(x_{u_{1i}}^t,y_{u_{1i}}^t,h_{u_{1i}}^t,...,x_{u_{\hat{M}i}}^t,y_{u_{\hat{M}i}}^t,h_{u_{\hat{M}i}}^t)$, along with the optimum position $\mathbf{g}_i^{t*}=\left(x_{u_{1i}}^{t*},y_{u_{1i}}^{t*},h_{u_{1i}}^{t*},...,x_{u_{\hat{M}i}}^{t*},y_{u_{\hat{M}i}}^{t*},h_{u_{\hat{M}i}}^{t*}\right)$ in each iteration t form a discrete state, denoted by $\sigma_i^t=(\mathbf{g_i}^t,\mathbf{g_i}^{t*})$, $i\in\{1,...,\Omega\}$. In addition, all possible states of ith member form a state space, denoted by Σ_i. Moreover, the set of all members' states is called the group state, denoted by $\sigma^t=(\sigma_1^t,...,\sigma_\Omega^t)$. With a similar argument as in (13), each state of the group σ^t depends only on its previous state, and accordingly, the group state σ^t is considered as a first-order Markov chain process. In addition, the set of all possible group statuses is denoted by Σ. Without loss of generality, we only consider the first phase for updating $\mathbf{g}_i^t$, expressed in (**??**). In the next step, we define the transition probability between successive states related to i^{th} member as follows:

$$\begin{aligned}\Pr\{\sigma_i^{t+1}=s_{t+1}|\sigma_i^t=s_t\}&=\Pr\{\mathbf{g}_i^t\rightarrow\mathbf{g}_i^{t+1}\}\times\Pr\{\mathbf{g}_i^{t*}\rightarrow\mathbf{g}_i^{t+1*}\}\\&=\frac{1}{\zeta}\times\frac{1}{\left\|\left(2-\frac{\mathcal{U}_n^i}{\mathcal{U}_n^{\text{best}}}\right)\times\mathbf{g}_i^t-\mathbf{g}_{\text{worst}}\right\|},\end{aligned}\tag{15}$$

where $\mathcal{U}^{update,i}\geq\mathcal{U}^i$. In addition, for the optimum position $\mathbf{g}_i^{t*}$, $\Pr\{\mathbf{g}_i^{t*}\rightarrow\mathbf{g}_i^{t+1*}\}=1$ if $\|\mathbf{g}_i^{t+1*}<\mathbf{g}_i^{t\,*}\|$, otherwise, $\Pr\{\mathbf{g}_i^{t\,*}\rightarrow\mathbf{g}_i^{t+1*}\}=0$. **Lemma 1:** The transition probability of Markov chain related to i^{th} member, denoted by $\Pr\{\sigma_i^{t+1}=s_{t+1}|\sigma_i^t=s_t\}$, is bounded as follows:

$$\begin{aligned}\prod_{i=1}^{\Omega}\Pr\{\sigma_i^{t+1}=s_{t+1}|\sigma_i^t=s_t\}-\kappa_1&\leq\Pr\{\sigma^{t+1}=s_{t+1}|\sigma^t=s_t\}\\&\leq\prod_{i=1}^{\Omega}\Pr\{\sigma_i^{t+1}=s_{t+1}|\sigma_i^t=s_t\}+\kappa_1.\end{aligned}\tag{16}$$

Proof. Defining $\Pr\{\sigma_i^{t+1}=s_{t+1}|\sigma_i^t=s_t\}=\mathrm{P}(X_i)$, we compute $\Pr\{\sigma^{t+1}=s_{t+1}|\sigma^t=s_t\}$ as follows:

$$\begin{aligned}\Pr\{\sigma^{t+1}=s_{t+1}|\sigma^t=s_t\}&=\mathrm{P}(X_1,X_2,...,X_\Omega)\\&=\mathrm{P}(X_1)\mathrm{P}(X_2|X_1)...\mathrm{P}(X_\Omega|X_1,...,X_{\Omega-1}).\end{aligned}\tag{17}$$

Since the exact probability of $\Pr\{\sigma^{t+1}=s_{t+1}|\sigma^t=s_t\}$ is difficult to compute, we determine the probability bounds instead of a closed-form expression for the probability, expressed as follows:

$$\begin{aligned}\mathrm{P}(X_1)(\mathrm{P}(X_2)-\kappa)...(\mathrm{P}(X_\Omega)-\kappa)&\leq\mathrm{P}(X_1,X_2,...,X_\Omega)\\&\leq\mathrm{P}(X_1)(\mathrm{P}(X_2)+\kappa)...(\mathrm{P}(X_\Omega)+\kappa),\end{aligned}\tag{18}$$

which comes from the expression $\mathrm{P}(B)-\kappa\leq\mathrm{P}(B|A)\leq\mathrm{P}(B)+\kappa$ [3]. Then, the probability bounds of $\mathrm{P}(X_1,X_2,...,X_\Omega)$ is re-expressed as:

$$\prod_{i=1}^{\Omega}\mathrm{P}(X_i)-\kappa_1\leq\mathrm{P}(X_1,X_2,...,X_\Omega)\leq\prod_{i=1}^{\Omega}\mathrm{P}(X_i)+\kappa_1,\tag{19}$$

or equivalently,

$$\begin{aligned}\prod_{i=1}^{\Omega}\Pr\{\sigma_i^{t+1}=s_{t+1}|\sigma_i^t=s_t\}-\kappa_1&\leq\Pr\{\sigma^{t+1}=s_{t+1}|\sigma^t=s_t\}\\&\leq\prod_{i=1}^{\Omega}\Pr\{\sigma_i^{t+1}=s_{t+1}|\sigma_i^t=s_t\}+\kappa_1\end{aligned}\tag{20}$$

In the next step, we define the super-martingale process in Definition 2 and prove that the proposed algorithm is a super-martingale process. Finally, we prove in Lemma 2 that the proposed machine-learning-based algorithm converges to a global optimal value.

Definition 2: Let assume $\{X^t|t\geq 0\}$ and $\{Y^t|t\geq 0\}$ as two stochastic process. Then, $\{X^t|t\geq 0\}$ is a super-martingale process with respect to $\{Y^t|t\geq 0\}$ if the following conditions are satisfied:

$$1)\ \mathbb{E}\left[X^{t+1}|Y^t=s_t,Y^{t-1}=s_{t-1},....,Y^0=s_0\right]\leq Y^t,\tag{21}$$

$$2)\ \mathbb{E}[X^t]<+\infty,\tag{22}$$

$$3)\ X^t\text{ is a function of }\{Y^0,...,Y^t\}.\tag{23}$$

Denoting $X^t = \mathcal{U}^t$ and $Y^t = \sigma^t$, we prove that $\mathcal{U}^t$ is a super-martingale process with respect to state σ^t, expressed as:

$$\begin{aligned}
&\mathbb{E}\left[\mathcal{U}\left(\sigma^{t+1}\right)|\sigma^t = s_t, \sigma^{t-1} = s_{t-1}, , ...\sigma^1 = s_1, \sigma^0 = s_0\right] \quad (24)\\
&\overset{(a)}{=} \mathbb{E}\left[\mathcal{U}\left(\sigma^{t+1}\right)|\sigma^t = s_t\right],\\
&= \sum_{s_{t+1}\in S_2} \mathcal{U}(\sigma^{t+1} = s_{t+1})\text{Pr}\{\sigma^{t+1} = s_{t+1}|\sigma^t = s_t\},\\
&\overset{(b)}{\leq} \sum_{s_{t+1}\in \Psi} \mathcal{U}(\sigma^t = s_t)\text{Pr}\{\sigma^{t+1} = s_{t+1}|\sigma^t = s_t\},\\
&\overset{(c)}{=} \mathcal{U}(\sigma^t = s_t),
\end{aligned}$$

where (a) comes from the Markov property of the process, (b) is based on the fact that the utility function is a minimization problem, i.e., $\mathcal{U}^{t+1}(s_{t+1}) \leq \mathcal{U}^t(s_t)$. In addition, (c) is hold due to the fact that Σ is a closed set. Accordingly, the first condition in Definition 2 is met. Moreover, $\mathbb{E}[\mathcal{U}^t]$ is upper-bounded as:

$$\begin{aligned}
\mathbb{E}[\mathcal{U}\left(\sigma^t\right)] &= \sum_{s_{t-1}\in\Sigma} \mathbb{E}[\mathcal{U}\left(\sigma^t\right)|\sigma^{t-1} = s_{t-1}] \quad (25)\\
&\leq \sum_{s_{t-1}\in\Sigma} \mathcal{U}(\sigma^t = s_{t-1}) < +\infty,
\end{aligned}$$

which satisfies the second condition in Definition 2. Thus, the proposed algorithm is a super-martingale process.

Lemma 2: Assuming X is a super-martingale process, there exists a random variable $\{\sigma^\infty \in \sigma^t | t > 0\}$ to which the population-based process converges with probability 1 when $t \to +\infty$. Proof: According to Lemma 1 and Definition 2, we conclude that there exists an optimal value $\{\sigma^\infty \in \sigma^t | t > 0\}$ to which the proposed algorithm converges. If σ^∞ belongs to the set of optimal states Σ^*, then, the global convergence can be guaranteed; otherwise, it does not exist a global optimum for the process. To prove the convergence of the algorithm, let assume $S_1 = \{\sigma^t = s_t, t \geq 1 | \sigma^t \notin \Sigma^*\}$, $S_2 = \{\sigma^t = s_t^*, t \geq 1\}$, where $S_2 \bigcap \Sigma^* \neq \Phi$, and $\delta = \min|\mathcal{U}\left(\sigma^t = s_t\right) - \mathcal{U}\left(\sigma^{t+1} = s_{t+1}^*\right)|$. In addition, there exists $r_1 \in \mathcal{R}^+$ such that the following constraint is satisfied:

$$\mathcal{U}\left(\sigma^t = s_t\right) - \mathcal{U}\left(\sigma^{t+1} = s_{t+1}^*\right) = r_1\delta. \quad (28)$$

Next, we compute $\mathbb{E}\left[\mathcal{U}\left(\sigma^t\right)\right] - \mathbb{E}\left[\mathcal{U}\left(\sigma^{t+1}\right)\right]$ as (26), where (a) comes from the the law of total expectations, i.e., $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X|Y]]$, where X and Y are random variables defined in the same space. In addition, (b) comes from the formula $\mathbb{E}[g(X)] = \sum_i g(X = x_i)\text{Pr}\{X = x_i\}$ for discrete random variable X. Then, $\mathbb{E}\left[\mathcal{U}\left(\sigma^t\right)\right] - \mathbb{E}\left[\mathcal{U}\left(\sigma^{t+1}\right)\right]$ is re-written as (27), where (a) comes from the formula (28). In addition, step (b) is related to the fact that S_1 is a subset of the set Σ. In the next step, we calculate $\sum_{t=1}^{N}\left(\mathbb{E}\left[\mathcal{U}\left(\sigma^t\right)\right] - \mathbb{E}\left[\mathcal{U}\left(\sigma^{t+1}\right)\right]\right)$, while $N \to +\infty$, expressed as:

$$\begin{aligned}
&\lim_{N\to+\infty}\sum_{t=1}^{N}\left(\mathbb{E}\left[\mathcal{U}\left(\sigma^t\right)\right] - \mathbb{E}\left[\mathcal{U}\left(\sigma^{t+1}\right)\right]\right) \quad (29)\\
&= P(S_1)\sum_{t=1}^{+\infty}\left(\sum_{s_{t+1}^*\in\Sigma}\text{Pr}\left\{\mathcal{U}\left(\sigma^{t+1} = s_{t+1}^*|\sigma^t = s_t\right)\right\}r_1\delta\right),\\
&\mathbb{E}\left[\mathcal{U}\left(\sigma^1\right)\right] - \mathbb{E}\left[\mathcal{U}\left(\sigma^{+\infty}\right)\right] \quad (30)\\
&= P(S_1)\sum_{t=1}^{+\infty}\left(\sum_{s_{t+1}^*\in\Sigma}\text{Pr}\left\{\mathcal{U}\left(\sigma^{t+1} = s_{t+1}^*|\sigma^t = s_t\right)\right\}r_1\delta\right).
\end{aligned}$$

Accordingly, if $P(S_1) \to 0$, then, $\sigma^{+\infty} \in \Sigma^*$, therefore, the algorithm converges to a global optimum. In addition, if $P(S_1) \to \eta, 0 \leq \eta \leq 1$; therefore,

$$\sum_{t=1}^{+\infty}\left(\sum_{s_{t+1}^*\in\Sigma}\text{Pr}\left\{\mathcal{U}\left(\sigma^{t+1} = s_{t+1}^*|\sigma^t = s_t\right)\right\}r_1\delta\right) < +\infty, \quad (31)$$

and all the states converge to a local optimum, and the proof is complete. □

Finally, we prove the convergence of Algorithm 2 related to subproblem $\mathscr{P}\mathbf{4}$ using Theorem 2.

Theorem 2: Let us denote the sequence generated by the iterative Algorithm 2 as $\{\boldsymbol{\nu}(\upsilon)\}_{\upsilon=1}^{\infty}$. Hence, any limit point $\bar{\boldsymbol{\nu}}^t$ and $\bar{\boldsymbol{\nu}}^e$ of $\{\boldsymbol{\nu}^t(\upsilon)\}_{\upsilon=1}^{\infty}$ and $\{\boldsymbol{\nu}^e(\upsilon)\}_{\upsilon=1}^{\infty}$ has the following features:

i) $\bar{\boldsymbol{\nu}}$ is a KKT point that minimizes (35),
ii) $f\left(\boldsymbol{\nu}^t(\upsilon), \boldsymbol{\nu}^e(\upsilon); \eta\right)$ converges monotonically to $f\left(\boldsymbol{\nu}^{t^*}, \boldsymbol{\nu}^{e^*}; \eta\right)$ for some KKT point $\boldsymbol{\nu}^{t^*}$ and $\boldsymbol{\nu}^{e^*}$.

Proof. Let $\boldsymbol{\nu}(\upsilon_k) \to \bar{\boldsymbol{\nu}}$ be one convergent sub-sequence so that

$$\lim_{k\to\infty}\boldsymbol{\nu}(\upsilon_k + 1) = \lim_{k\to\infty}\boldsymbol{\nu}(\upsilon_k) = \bar{\boldsymbol{\nu}}. \quad (32)$$

Since,

$$\boldsymbol{\nu}(\upsilon_k + 1) := \arg\min_{\boldsymbol{\nu}\in\boldsymbol{\mathcal{Z}}}\mathcal{D}\left(\boldsymbol{\nu}; \xi(\upsilon_k)\right), \quad (33)$$

which is a convex optimization problem, the KKT condition holds at $\boldsymbol{\nu}(\upsilon_k + 1)$, i.e.,

$$0 \in \nabla_{\boldsymbol{\nu}}\mathcal{D}\left(\boldsymbol{\nu}(\upsilon_k + 1); \xi(\upsilon_k)\right) + \mathcal{L}_{\mathcal{Z}}\left(\boldsymbol{\nu}(\upsilon_k + 1)\right). \quad (34)$$

According to [4, Proposition 6.6], and (32), we have

$$\lim_{\boldsymbol{\nu}(\upsilon_k+1)\to\bar{\boldsymbol{\nu}}}\sup \quad \mathcal{L}_{\mathcal{Z}}\left(\boldsymbol{\nu}(\upsilon_k + 1)\right) = \mathcal{L}_{\mathcal{Z}}\left(\bar{\boldsymbol{\nu}}\right). \quad (35)$$

Furthermore, using (32), we have

$$\begin{aligned}
\lim_{k\to\infty}\nabla_{\boldsymbol{\nu}}\mathcal{D}\left(\boldsymbol{\nu}(\upsilon_k + 1); \xi(\upsilon_k)\right) &= \lim_{k\to\infty} 2\sum_{i=1}^{m}\xi(\upsilon_k)\boldsymbol{\nu}(\upsilon_k + 1)\\
= \lim_{k\to\infty}\sum_{i=1}^{m}\frac{p\,\boldsymbol{\nu}(\upsilon_k + 1)}{\left[\left(\boldsymbol{\nu}(\upsilon_k)^2 + \xi^2\right)\right]^{1-\frac{p}{2}}} &= \nabla_\nu f\left(\bar{\boldsymbol{\nu}}; \eta\right) \quad (36)
\end{aligned}$$

Therefore, taking $k \to \infty$ in (34), we obtain

$$0 \in \nabla_{\boldsymbol{\nu}}\mathcal{D}\left(\bar{\boldsymbol{\nu}}; \bar{\xi}\right) + \mathcal{L}_{\mathcal{Z}}\left(\bar{\boldsymbol{\nu}}\right), \quad (37)$$

which indicates that $\bar{\boldsymbol{\nu}}$ is a KKT point that minimizes (35), and this completes the proof of (i).

$$\mathbb{E}\left[\mathcal{U}\left(\sigma^{t}\right)\right]-\mathbb{E}\left[\mathcal{U}\left(\sigma^{t+1}\right)\right] \overset{(a)}{=} \mathbb{E}\left[\mathcal{U}\left(\sigma^{t}\right)\right]-\mathbb{E}\left[\mathbb{E}\left[\mathcal{U}\left(\sigma^{t+1}\right)|\mathcal{U}\left(\sigma^{t}\right)\right]\right] \overset{(b)}{=} \sum_{s_t\in\Sigma}\Pr\left\{\sigma^{t}=s_t\right\}\left(\mathcal{U}\left(\sigma^{t}=s_t\right)-\mathbb{E}\left[\mathcal{U}\left(\sigma^{t+1}\right)|\mathcal{U}\left(\sigma^{t}=s_t\right)\right]\right)$$
$$=\sum_{s_t\in\Sigma}\Pr\left\{\sigma^{t}=s_t\right\}\left(\mathcal{U}\left(\sigma^{t}=s_t\right)-\sum_{s_{t+1}^{*}\in\Sigma}\mathcal{U}\left(\sigma^{t+1}=s_{t+1}^{*}\right)\Pr\left\{\mathcal{U}\left(\sigma^{t+1}=s_{t+1}^{*}\right)|\mathcal{U}\left(\sigma^{t}=s_t\right)\right\}\right), \quad (26)$$

$$\begin{aligned}\mathbb{E}\left[\mathcal{U}\left(\sigma^{t}\right)\right]-\mathbb{E}\left[\mathcal{U}\left(\sigma^{t+1}\right)\right] &= \sum_{s_t\in\Sigma}\Pr\left\{\sigma^{t}=s_t\right\}\left(\mathcal{U}\left(\sigma^{t}=s_t\right)\sum_{s_{t+1}^{*}\in\Sigma}\Pr\left\{\mathcal{U}\left(\sigma^{t+1}=s_{t+1}^{*}\right)|\mathcal{U}\left(\sigma^{t}=s_t\right)\right\}\right. \\ &\quad \left.-\sum_{s_{t+1}^{*}\in\Sigma}\mathcal{U}\left(\sigma^{t+1}=s_{t+1}^{*}\right)\Pr\left\{\mathcal{U}\left(\sigma^{t+1}=s_{t+1}^{*}\right)|\mathcal{U}\left(\sigma^{t}=s_t\right)\right\}\right) \\ &\overset{(a)}{=} \sum_{s_t\in\Sigma}\Pr\left\{\sigma^{t}=s_t\right\}\sum_{s_{t+1}^{*}\in\Sigma}\Pr\left\{\mathcal{U}\left(\sigma^{t+1}=s_{t+1}^{*}|\sigma^{t}=s_t\right)\right\}r_1\delta \\ &\overset{(b)}{\geq} \sum_{s_t\in S_1}\Pr\left\{\sigma^{t}=s_t\right\}\sum_{s_{t+1}^{*}\in\Sigma}\Pr\left\{\mathcal{U}\left(\sigma^{t+1}=s_{t+1}^{*}|\sigma^{t}=s_t\right)\right\}r_1\delta \\ &= \sum_{s_{t+1}^{*}\in\Sigma}\Pr\left\{\mathcal{U}\left(\sigma^{t+1}=s_{t+1}^{*}|\sigma^{t}=s_t\right)\right\}r_1\delta P(S_1),\end{aligned} \quad (27)$$

Since $f\left(\boldsymbol{\nu}\,;\eta\right)$ is continuous and $\mathcal{Z}$ is compact, we can claim that the limit of the sequence $f\left(\boldsymbol{\nu}\left(\upsilon\right);\eta\right)$ is finite. Moreover, we have $f\left(\boldsymbol{\nu}\left(\upsilon+1\right);\eta\right)\leqslant f\left(\boldsymbol{\nu}\left(\upsilon\right);\eta\right)$ based on (46). According to the results obtained in the step (i), the proof of (ii) is completed. □

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